CSD206 Computer Organization - Class Assignment 2

Instructions

- This assignment is based on the research papers uploaded on September 30, 2018.
- **Assignment submitted after due date and time will not be evaluated and a score of zero will be awarded for this assignment.**
- **Since it's a class assignment, you are required to be present during the lecture timings, else your submission will not be considered for evaluation.**
- Materials copied from the Internet or otherwise will attract penalty as per the course policy.
- Upload a **pdf version** of the document.

Due Date: 11:55 am, Oct 4, 2018

Submitting this Assignment

You will submit (upload) this assignment on Blackboard. Email submissions will not be accepted.

- Write your answer after the given question in this document.
- Name this document as CA2_CO2018_John_Doe.pdf in case your name is John Doe.

Grading: This assignment has 2 points (with **weightage of 2%** in your overall 100 points)

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Answer each of the following question in about 400 – 500 words.

1. **Write in detail the characteristics of RISC instruction set.**

Answer:

One of the most significant recent developments in computer processor technology is the RISC microprocessor. RISC stands for Reduced Instruction Set Computer, this was a new technology that first emerged in the 1980s as an attempt to make more computing more efficient by reducing the number of instructions, addressing modes and formats. They trade off program length, to minimize time to execute instruction.

An ideal RISC has the following characteristics:

- It has simple instructions, supports few simple data types, and helps synthesize complex data types.
- RISC uses Harvard Architecture.
- Most instructions have the same size and execute within a single clock cycle.
- The architecture has a large number of general-purpose registers. (16 to 32)
- It has hard-coded logic (hard-wired control unit).
- There are very limited addressing modes, and instructions are conceptually simple.
- It has fixed-length instructions that do not span word boundaries.
- They have pipelined instruction cycles, which are typically highly efficient uniform delay pipelines.
- A compiler is used to perform the conversion operation means to convert a high-level language statement into the code of its form.
- Only LOAD/STORE instructions access memory.
- All operate instructions are register-to-register instructions, and so, RISC uses a complicated register address decoding method.
- The average RISC program is about 30% longer than the average CISC program, and hence take up more memory.
- There is no floating point operation support in RISC, yet.
- The instruction set is orthogonal, i.e., there is little overlapping of instruction functionality.
- More RAM is required to store assembly level instructions.

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2. Compare and contrast RISC and CISC instruction set.

Feature	RISC	CISC
Full Form	Reduced Instruction Set Computer.	Complex Instruction Set Computer.
Registers	16-32 general purpose	8-16 general purpose
Data Types	Usually *two (*integer and floating point)	Usually more than *two.*
Instructions	Load/store general registers, operations on datatypes in registers.	Correspond to data types.
Instruction Formats	Instructions are only LOAD/STORE instructions, and have a fixed length that do not span word boundaries.	Instructions are of many types, and have a variable length, that span word boundaries.
Instruction execution	Faster	Slower
Instruction Time	Instructions take fixed amount of time.	Instructions take variable amount of time.
Instruction Set	The instruction set is reduced i.e. it has only a few instruction in the instruction set. Many of these instructions are very primitive.	The instruction set has a variety of different instructions that can be used for complex operations.
Number of Instructions	The same program in RISC often requires many more instructions than CISC.	The same program in CISC often requires much fewer instructions than RISC.
Addressing Modes	Very limited. Complex addressing modes are synthesized using the software.	Extensive and highly sophisticated addressing modes already exist.
Average CPI	1.5	Between 2 and 15.
Encoding	An instruction contains only a single operand or operation	An instruction contains an entire statement.
Design Objective	They trade off program length, to minimize time to execute instruction.	They trade off work per instruction, to minimize program length.
Processor	Hard-wired processor	Microprogrammed processor
Pipeline	Simple and highly efficient pipeline.	Complex and not as efficient Pipeline.
Pipelining branch penalties	Less.	More.
Caching	Essential for instructions.	Useful for instructions.

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Cache Performance	Good	Better
Compiler Design	Should stress best ordering.	Should stress finding right instructions.
Philosophy	Move all functions to software.	Move any useful software function into hardware, including diagnostics.
Optimization	Performance optimized with more focus of software.	Performance optimized with more focus of hardware.
Memory Requirement	More	Less
Control Unit	Simple, since it is dealing with less instructions and addressing modes.	Complex.
Decoding	Decoding is faster, since it is much easier in RISC due to limited instructions and addressing modes, and fixed-size instructions.	Decoding is slower.
R&D	Faster processor design, development and test, due to a simpler design.	Slower process.
Context Switching	Slower	Faster
Compiler Design Requirements	More	Less
Bus Traffic	High	Reduced
Scaling Ability	Better	Worse
Chip Area Usage	Better	Worse